



PIONEER JUNIOR COLLEGE
JC2 PRELIMS EXAMINATIONS 2015

**Candidate
Name**

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**Civics
Group**

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**INDEX
NUMBER**

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H2 BIOLOGY

9648 / 01

23 September 2015

1 hour 15 mins

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number on the cover page in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the OTAS answer sheet provided.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Do not open this booklet until you are told to do so.

This document consists of **27** printed pages (inclusive of this page)

1. The diagram shows a section of a cell. Which of the following correctly matches the structure of the labelled component to its function?

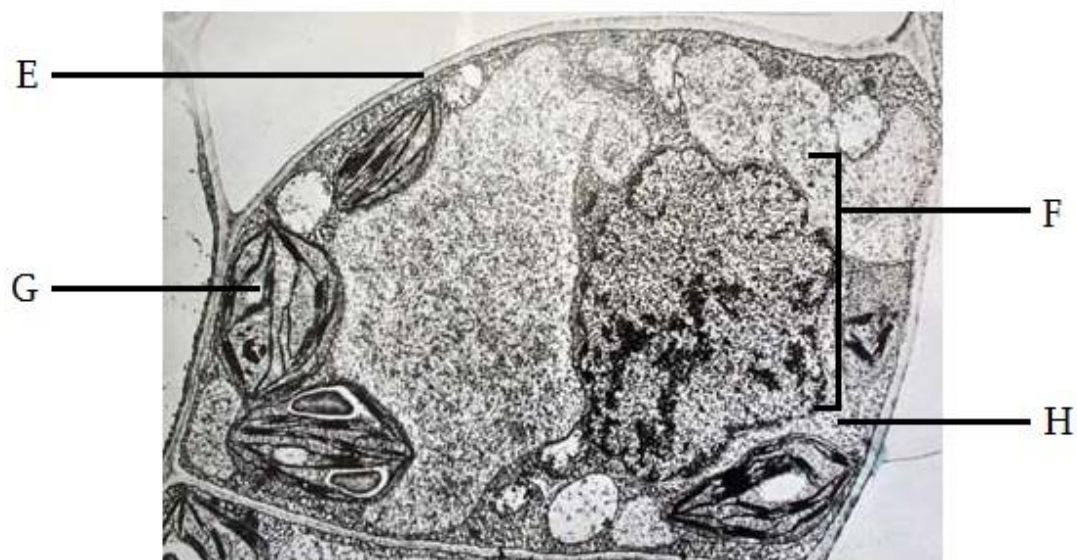


Image adapted from: http://umanitoba.ca/Biology/BIO11020/lab3/lab3_2.htm

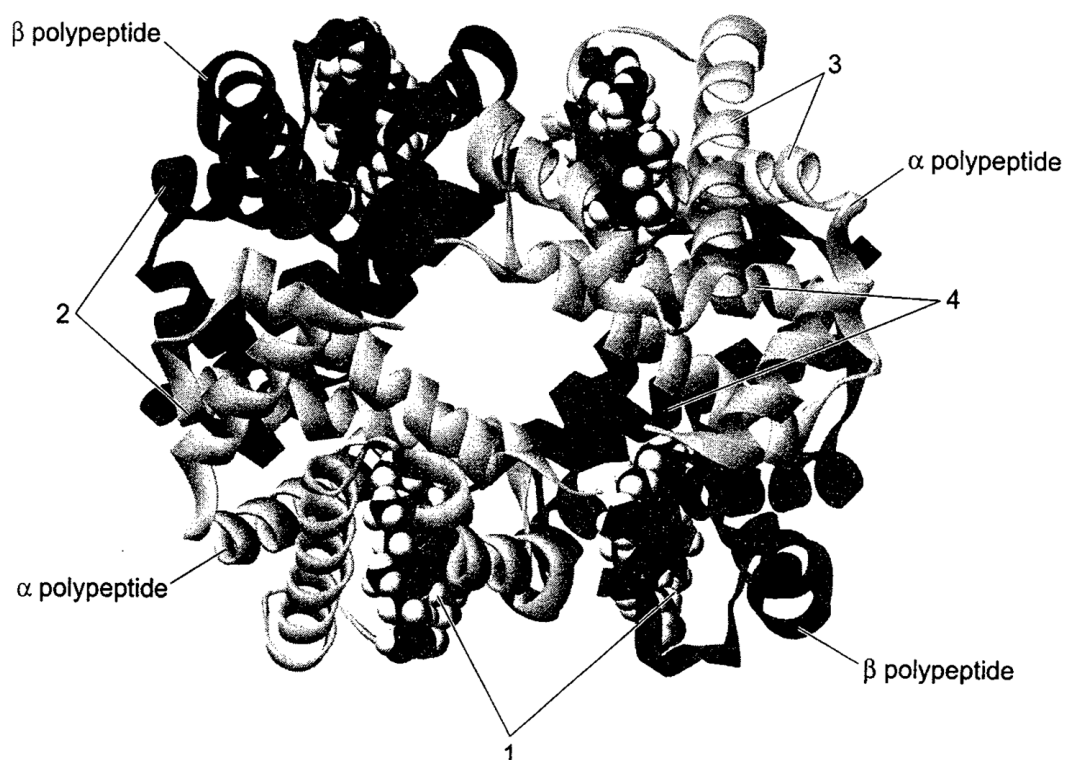
	Component	Structure	Function
I	E	Extensive hydrogen bonds between tropocollagen molecules	High tensile strength to maintain cell shape
II	F	Selectively permeable due to its phospholipid bilayer	Regulate entry and exit of ribosomal proteins
III	G	ATP synthase embedded in inner membrane	Synthesises ATP for Krebs cycle
IV	H	Presence of glycolytic enzymes	Oxidises glucose to pyruvate

- A II only
 B **II and IV only**
 C II, III and IV
 D all of the above

2. Which statement best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?

- A** Presence or absence of 180° rotation of alternate subunits
- B Presence or absence of branching in the macromolecule
- C Subunits are either monosaccharides or amino acids
- D Different tendency of macromolecules to form hydrogen bond with water

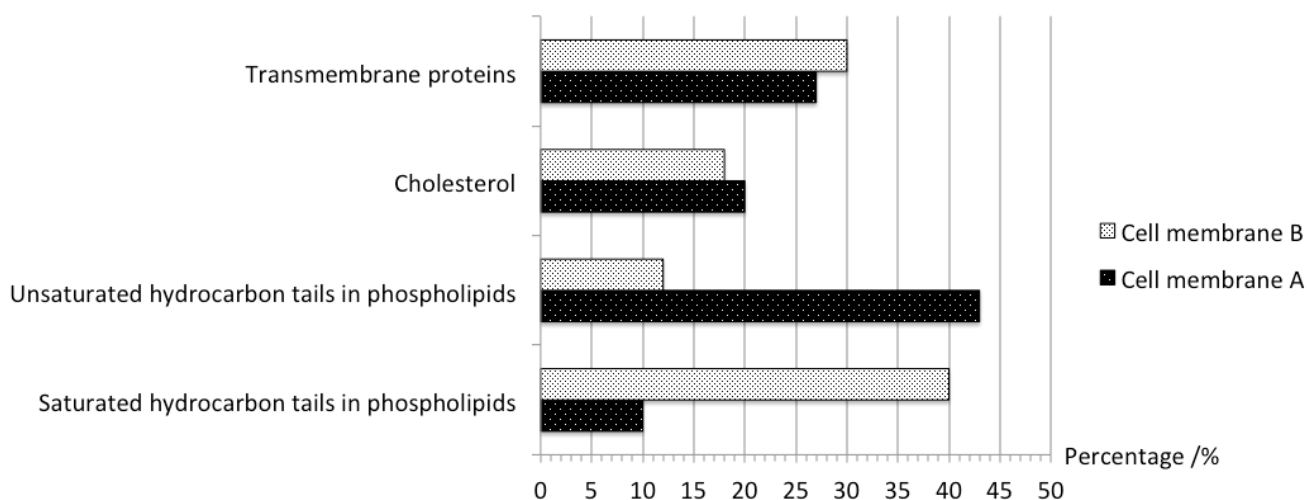
3. The diagram shows a haemoglobin molecule.



Which identifies the different parts of the molecule?

	1	2	3	4
A	prosthetic group	beta pleated sheet	alpha helix	hydrophilic amino acid
B	hydrophobic amino acids	beta pleated sheet	prosthetic group	binding site
C	prosthetic group	hydrophilic amino acids	alpha helix	hydrophobic amino acids
D	prosthetic group	hydrophilic amino acid	binding site	hydrophobic amino acids

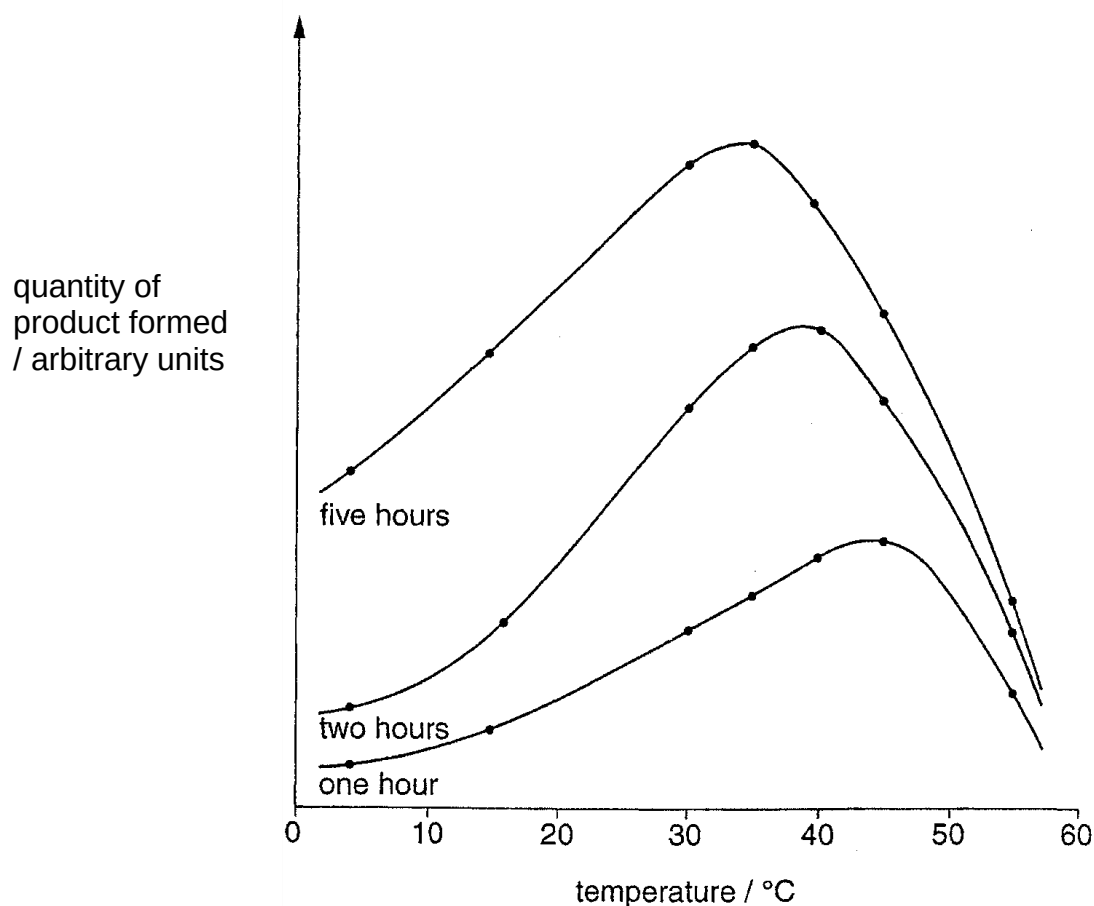
4. Reindeer are well adapted to survive extreme cold winters. One of these adaptations is the cell membrane composition at different parts of its body. The graph below shows the percentage composition of cell membrane components of cells A and B taken from two different parts of the reindeer's body.



Which of the following statement best explains the differences in the membrane composition in Cell A and Cell B?

- A Cholesterol decreases the membrane fluidity and prevents the membrane from breaking up by restraining the movement of phospholipids.
- B Cell membrane A is taken from a lower part of the reindeer's leg as the unsaturated hydrocarbon tails will prevent the fatty acids from packing close to each other.**
- C Cell membrane B is taken from a lower part of the reindeer's leg as the saturated hydrocarbon tails will prevent the fatty acids from packing close to each other.
- D Transmembrane proteins maintain the osmotic balance between the interior and exterior of the cell, hence preventing the cell membrane from solidifying at low temperatures.

5. The following figure shows the results of an experiment in which samples containing the same concentration of enzyme and substrate were kept at different temperatures for periods of one, two and five hours. The quantities of product formed were then determined.



Which statement below does NOT describe the graphs?

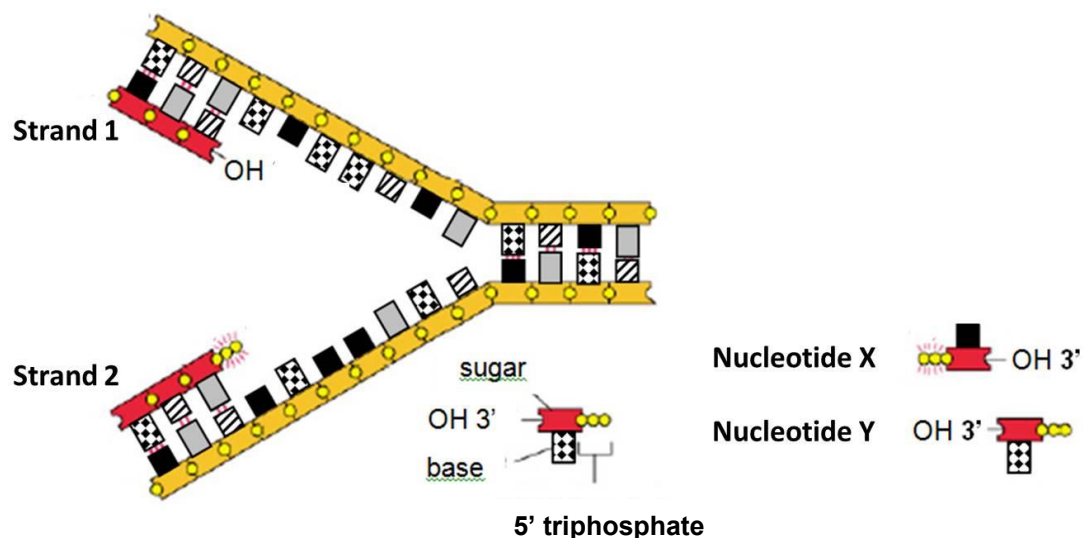
- A** Optimum temperature for the experiment held over one hour was higher as the enzyme had more disulfide bonds in stabilizing its structure.
- B** As temperature increased, the quantity of products formed increased.
- C** As the duration of the experiment decreased, the smaller the quantity of products formed.
- D** As the duration of the experiment increased, the rate of denaturation of the enzyme becomes faster.

6. Let the amount of DNA present in the nucleus of a diploid cell before interphase be represented by 2X picograms (pg).

Which of the following combination reflects the amount of DNA and ploidy number in the nucleus during the various stages of cell division?

		Before interphase	After S phase of interphase	After telophase of meiosis I	After prophase of meiosis II	After telophase of meiosis II
A	Amount of DNA /pg	2X	4X	2X	4X	2X
	Ploidy number	2n	2n	2n	2n	n
B	Amount of DNA /pg	2X	4X	2X	2X	X
	Ploidy number	2n	2n	n	n	n
C	Amount of DNA /pg	2X	4X	2X	2X	X
	Ploidy number	2n	2n	2n	2n	n
D	Amount of DNA /pg	2X	4X	2X	4X	2X
	Ploidy number	2n	2n	n	n	n

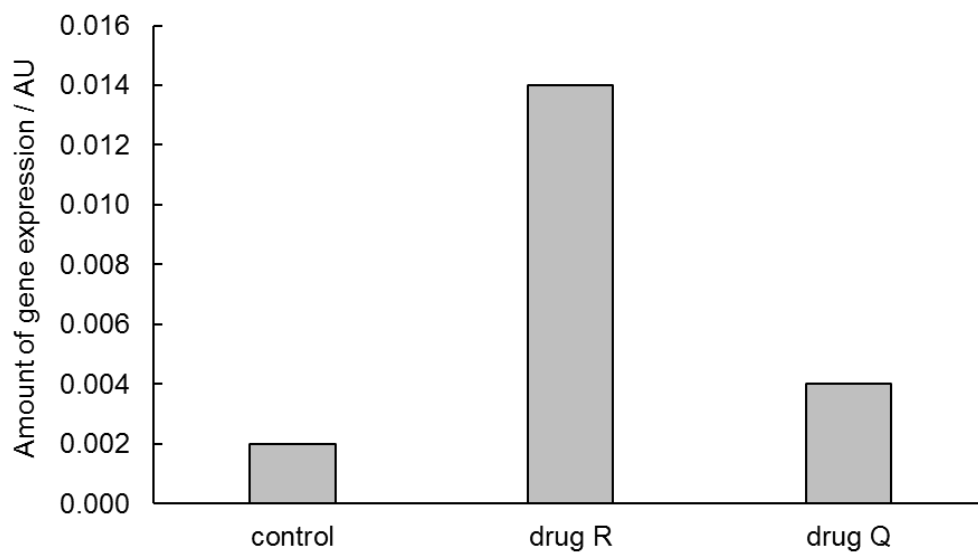
7. DNA replication is illustrated in the following figure.



Which of the following correctly describes the addition of the next nucleotide(s) in the DNA strands undergoing replication?

- A** Nucleotide X will be added to the leading strand, which is strand 1.
- B** Nucleotide Y will be added to the leading strand, which is strand 1.
- C** Nucleotide X will be added to the lagging strand, which is strand 1.
- D** Nucleotide Y will be added to the leading strand, which is strand 2.

8. Drug R is a DNA methyltransferase inhibitor and drug Q is a histone deacetylase inhibitor. An experiment was carried out to investigate the effects of drugs R and Q on expression of a gene. The graph shows the experimental results.



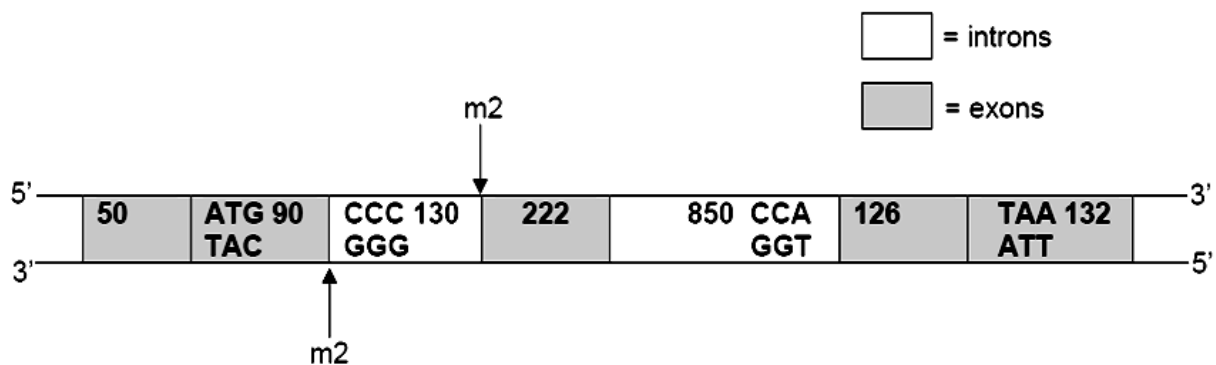
Which are possible explanations to the results shown?

- I. Drug Q results in weaker binding of histones to DNA.
- II. Drug Q increases gene expression by increasing accessibility of RNA polymerase to the promoter.
- III. Drug R increases gene expression by preventing methylation at CpG islands at the promoter.
- IV. Inhibiting DNA methylation is more effective in increasing gene expression than inhibiting histone deacetylation.

- A I and II only
- B II and IV only
- C I, II and III only
- D All of the above**

Use the information below to answer Questions 9 and 10.

The figure below shows the genomic structure of the wild-type human β -globin gene. The numbers within the boxes indicate the length of nucleotides of each region, **inclusive** of bases stated in the diagram. The DNA sequences corresponding to the start codon and the stop codon are indicated.



9. What is the length (in nucleotides) of the wild-type β -globin primary RNA transcript (pre-mRNA) and how many amino acids are present in the wild type β -globin protein?

	Length of β -globin primary RNA transcript	No. of amino acid in wild type β -globin protein
A	1600	146
B	620	146
C	1600	206
D	620	206

10. A base-pair substitution mutation (m2) indicated above, destroys the 3' splice site located in the first intron of the β -globin gene. The splice site is the site where the DNA will be cut by enzymes.

Choose the correct explanation as to whether the m2 mutation would result in any change in the length of the primary RNA transcript and mature RNA transcript made from the mutant β -globin gene.

	Pre-mRNA	Mature mRNA
A	No change as this is before the mRNA is spliced.	The splice site is damaged, the intron remains and the mature mRNA is increased by 130 bases.
B	No change as this is before the mRNA is spliced.	The splice site is damaged, the exon remains and the mature mRNA is increased by 222 bases.
C	An increased in its length as this is before RNA splicing.	The splice site is damaged, the intron remains and the mature mRNA is increased by 130 bases.
D	A decrease in its length as this is before RNA splicing.	The splice site is damaged, the exon remains and the mature mRNA is increased by 222 bases.

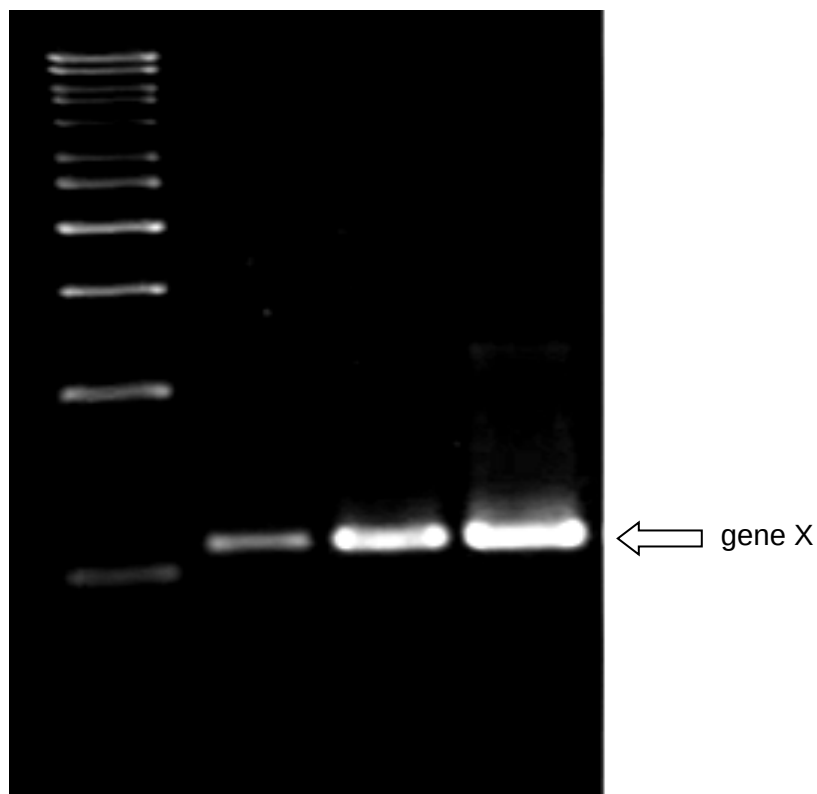
11. Some bacterial drugs can affect the synthesis of proteins.

Antimicrobial drug	Rifampicin	Streptomycin	Tetracycline
Mode of action	Binds to RNA polymerase	Genetic code misread during translation	Prevents binding of tRNA to ribosome

Which is the correct set of immediate effects of these drugs?

Antimicrobial drug	Rifampicin	Streptomycin	Tetracycline
A	Defective protein synthesized	mRNA does not bind to ribosome	Amino acids not added to growing chain
B	Transcription prevented	Defective protein synthesized	mRNA does not bind to ribosome
C	Defective protein synthesized	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesized	Amino acids not added to growing chain

12. Gel electrophoresis was performed using samples of gene X isolated from equal number of cells from a human embryo after 2 days, 4 days and 8 days of development.



DNA ladder	2 days	4 days	8 days
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What kind of gene regulation is illustrated by the results of gel electrophoresis?

- A DNA methylation
- B histone acetylation
- C gene amplification**
- D transcriptional activation

13. The table describes different kinds of mutations in DNA.

Mutation	Name
From purine to other purine	Transition
From pyrimidine to other pyrimidine	Transition
From purine to pyrimidine	Transversion
From pyrimidine to purine	Transversion

The diagram represents part of a DNA molecule.

G A T A C C A
C T A T G G T

Which diagram shows the DNA molecule with only transversion(s)?

- A

G T T A T C A
C A A T A G T
- B**

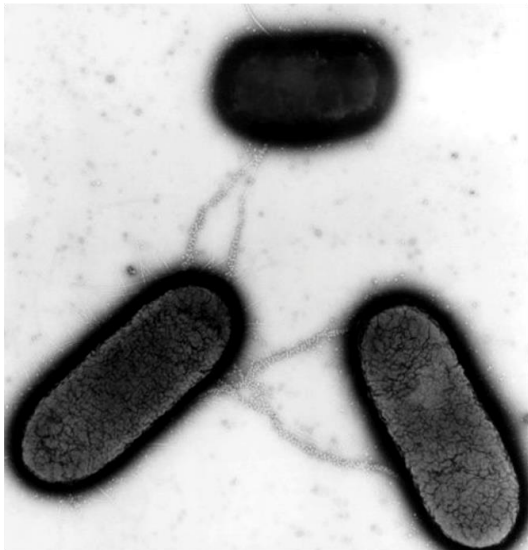
G A A A C A A
C T T T G T T
- C

A A T A C C A
T T A T G G T
- D

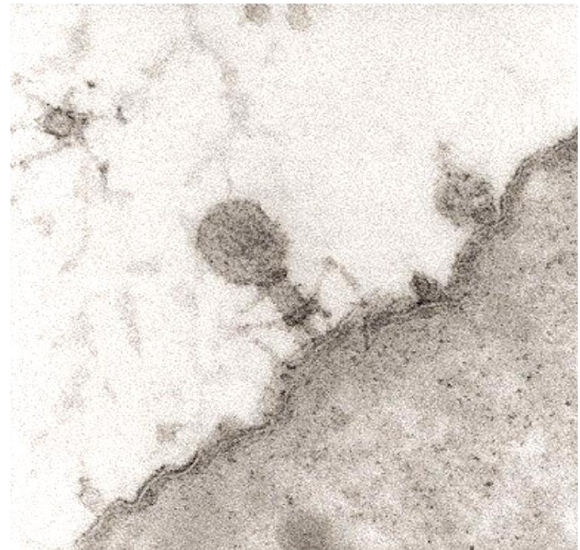
G A T A T C A
C T A T A G T

14. The photomicrographs below show two different processes of genetic transfer in bacteria.

Process R



Process S



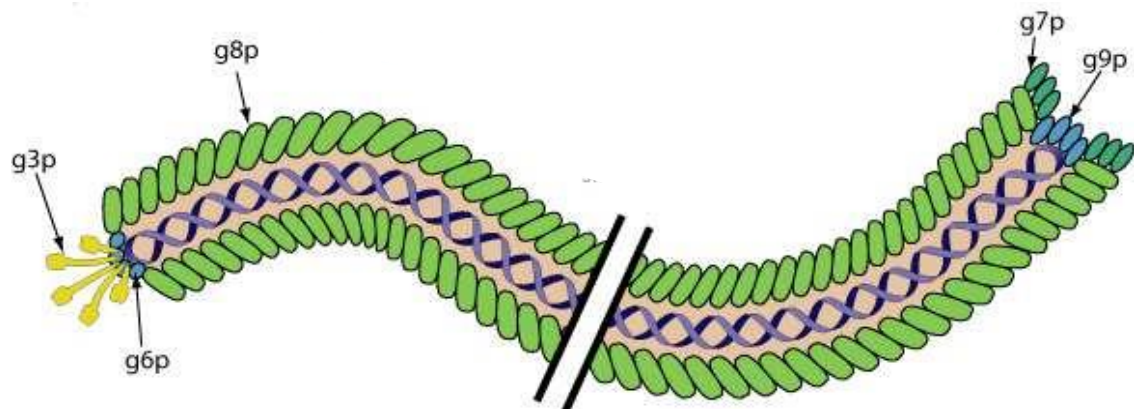
A series of statements concerning the two processes is shown below.

- I. Physical contact between the donor and recipient bacterium is necessary.
- II. The donor bacterium is lysed in the process.
- III. Specific genes in the donor bacterium are required for the transfer to take place.
- IV. Any gene on the bacterial chromosome can be transferred.

Which row shows the correct statements for the corresponding process?

	Process R	Process S
A	I and II	III and IV
B	I and III	II and IV
C	II and III	I and IV
D	III and IV	I and II

15. The diagram below shows the structure of M13 bacteriophage. It has single-stranded circular DNA genome. g3p, g6p, g7p, g8p and g9p are capsid proteins. Its life cycle does not cause the lysis of the host cell.

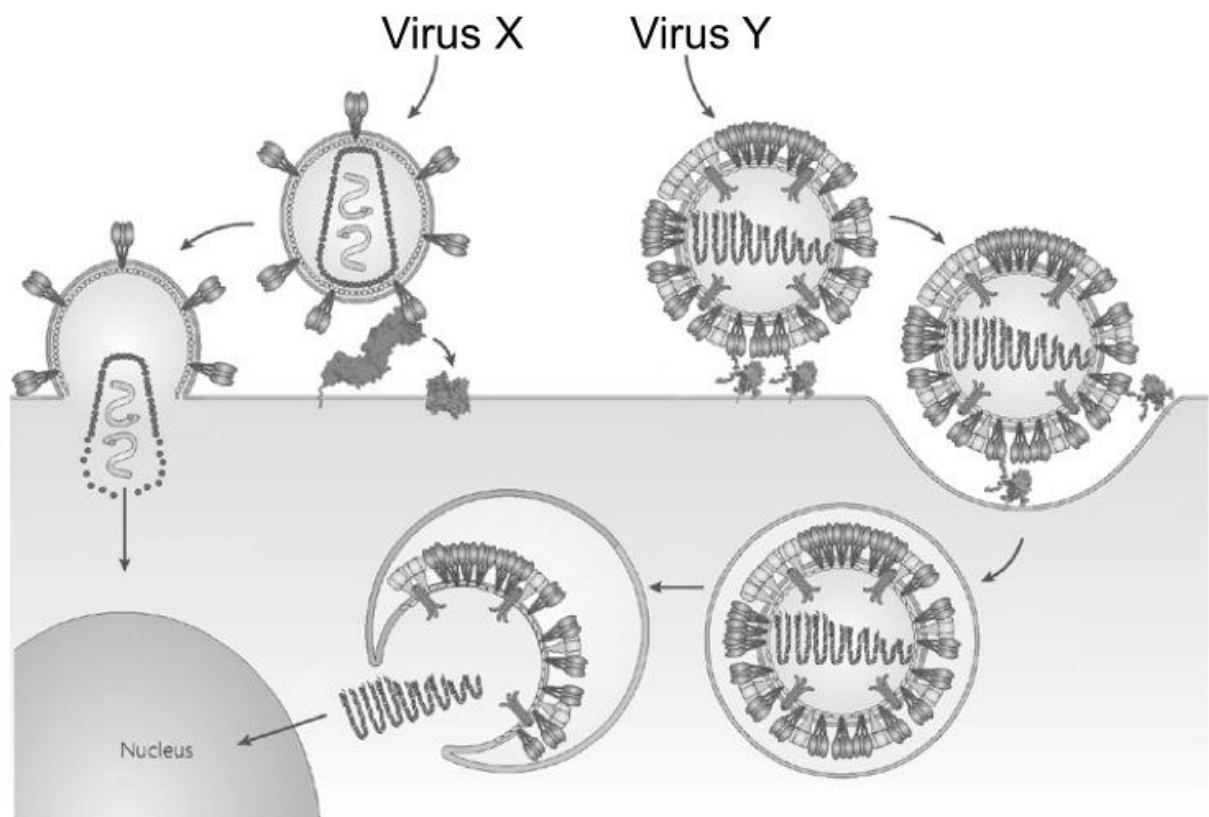


Based on your understanding of bacteriophages, which of the following statements are true of the M13 bacteriophage?

- I. The base composition of its genome is such that the relative ratio of C:G is 1:1.
- II. At least one of the capsid proteins is responsible for binding to a specific protein on the host cell.
- III. Its genome is injected into the host cell after the phage has attached to the host cell.
- IV. It acquires its envelope from the cell membrane of its host cell.

- A I and III
- B II and III**
- C I, II and III
- D II, III and IV

16. The diagram shows the entry of two types of animal viruses X and Y into host cells.



Which statements are true about the entry of viruses X and Y?

- I. For X, the viral envelope fuses with the plasma membrane of the host cells to allow entry of the virion into the host cell.
- II. For X, entry into the host cell is not receptor-mediated.
- III. For Y, the viral envelope fuses with the endosomal membrane to release the viral genetic material.
- IV. For Y, the endosomal membrane fuses with the plasma membrane to release the viral genetic material.

- A I, II and III
- B I and III**
- C II and IV
- D I, III and IV

17. Possession of white or coloured in poultry is controlled by two genes P / p and Q / q. The phenotypes of offspring that are expected from mating two birds, each of which is heterozygous at both loci, are shown in the Punnett square.

Gametes	PQ	Pq	pQ	pq
PQ	White feathers	White feathers	White feathers	White feathers
Pq	White feathers	White feathers	White feathers	White feathers
pQ	White feathers	White feathers	Coloured feathers	Coloured feathers
pq	White feathers	White feathers	Coloured feathers	White feathers

What best explains the proportion of white to coloured feathers in the Punnett square?

- A** dominant epistasis in which the epistatic allele is P
- B** dominant epistasis in which the epistatic allele is Q
- C** recessive epistasis in which the epistatic allele is p
- D** recessive epistasis in which the epistatic allele is q

18. A female drosophila with red eyes and normal wings was crossed with a male pure breeding for purple eyes and vestigial wings. The table shows the phenotypes of the offspring.

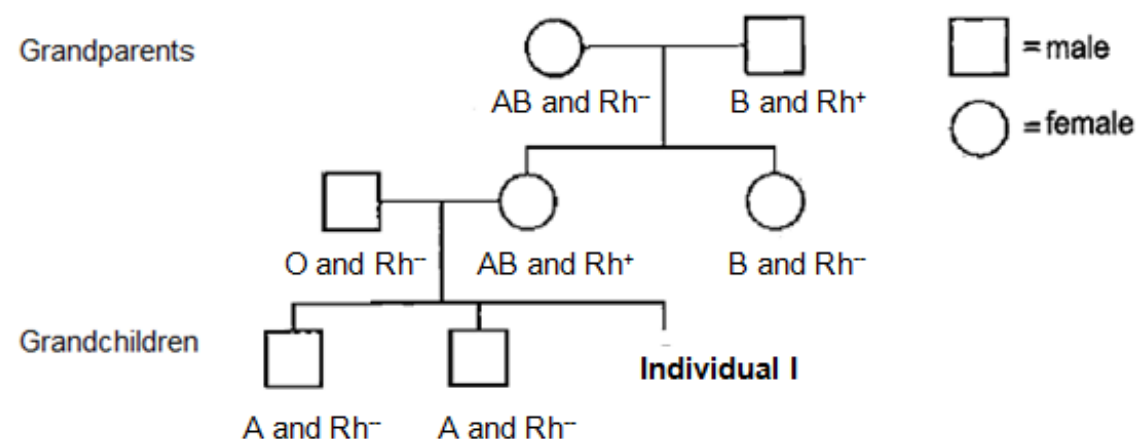
Red eye, normal wing	1139
Purple eye, vestigial wing	1195
Red eye, vestigial wing	151
Purple eye, normal wing	155

Which statement best explains these results?

- A** The genes for eye colour and wing type are on different chromosomes and 5.7 % of the offspring express two dominant alleles.
- B** The genes for eye colour and wing type are on different chromosomes and 5.9 % of the offspring express two recessive alleles.
- C** The genes for eye colour and wing type are linked and crossing over has taken place in an average of 5.8 % of the offspring.
- D** The genes for eye colour and wing type are linked and the distance between two genes are 11.6 centiMorgan.

19. Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?
- A Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.
 - B Identical genetic crosses performed between varieties of *Dendranthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glass house.
 - C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.
 - D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0 – 71.0 cm.

20. The diagram shows a family tree.



What is the probability that **Individual I** will be a girl with blood group B and Rh negative?

- A 1 in 16
- B 1 in 8
- C 1 in 4
- D 1 in 2

21. A plant researcher tried to investigate a cross between two heterozygous Snapdragon plants that produced red flowers. She predicted three possible phenotypic outcomes, namely plants with white flowers, pink flowers and red flowers, with a phenotypic ratio of 4:3:9 respectively. When the cross was performed, she found 50 plants with white flowers only, 41 plants with pink flowers, and 85 plants with red flowers. A chi-squared test was performed, and the chi-squared value was calculated to be 4.74.

Degree of freedom	Probability, P				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is correct?

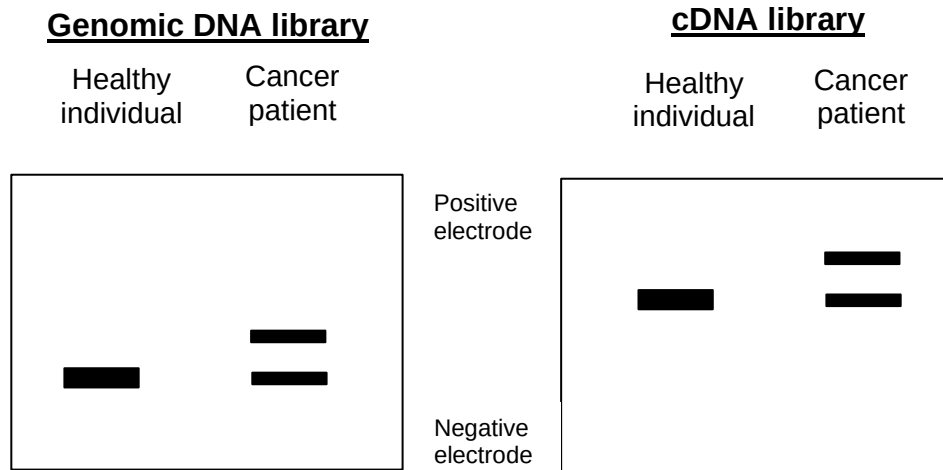
- A The degree of freedom is 3.
- B The calculated chi-squared value is greater than the critical chi-squared value.
- C There is a high probability that the difference between the observed and expected values is due to chance.
- D The probability that the difference between observed and expected values is due to chance is less than 5%.

22. Which statements about cancer are correct?

- I. Exposure to UV light from the sun always causes skin cancer.
- II. Some viruses can cause cancer.
- III. The mutation of genes controlling the cell cycle may cause cancer.
- IV. Inhaling too much carbon monoxide increases the risk of developing cancer.

- A I and II
- B I and III
- C II and III
- D II, III and IV

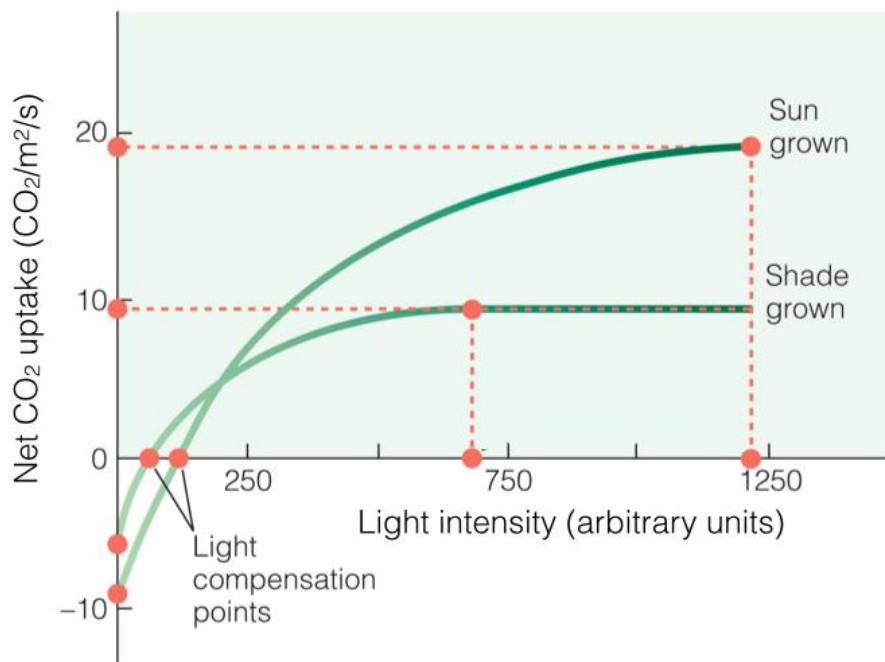
23. The genomic DNA and cDNA libraries of a cancer patient and a healthy individual were created to study the biology of liver cancer. A gene involved in the development of cancer was extracted from both libraries and studied.



Base on the results, which kind of gene was studied and what is the size of the mutant allele with respect to the normal allele?

	Gene studied	Size of mutant allele relative to normal allele
A	Proto-oncogene / oncogene	Mutant allele > normal allele
B	Proto-oncogene / oncogene	Mutant allele < normal allele
C	Tumour suppressor gene	Mutant allele > normal allele
D	Tumour suppressor gene	Mutant allele < normal allele

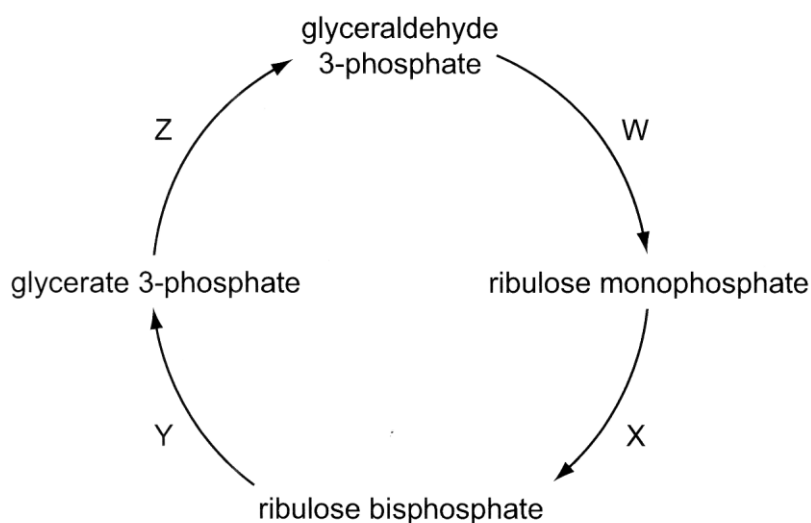
24. The effect of light intensity on photosynthetic rate was investigated in sun-grown and shade-grown leaves. The results obtained from this investigation are shown in the following graph.



Which of the following statement can be concluded?

- A** There are more palisade mesophyll cells in sun-grown leaves than shade-grown leaves, thus light saturation point for sun-grown leaves is higher.
- B** Shade-grown leaves are more efficient at harnessing light energy at high light intensity.
- C** Light compensation point of sun-grown leaves is higher than shade-grown leaves as sun-grown leaves require less carbon dioxide to carry out photosynthesis.
- D** Rate of Calvin cycle is faster in sun-grown leaves than shade-grown leaves at very low light intensity.

25. The diagram shows the main stages in the Calvin cycle.



At which stages are ATP and reduced NADP used and carbon dioxide taken up?

	ATP used	reduced NADP used	carbon dioxide taken up
A	W and Z	X	Y
B	X	W and Y	Z
C	Y	X and Z	W
D	X and Z	Z	Y

26. Which row correctly shows the reactions at the various locations in an eukaryotic cell?

	Cytoplasm	Mitochondrial matrix	Inner mitochondrial membrane
A	Phosphorylation of glucose 6-phosphate to glucose 1,6-bisphosphate	Oxidation of malate to oxaloacetate	Flow of electrons down ETC
B	Phosphorylation of fructose 6-phosphate to fructose 1,6-bisphosphate	Reduction of succinate to fumarate	Oxidative phosphorylation
C	Formation of ATP via substrate-level phosphorylation	Reduction of malate to oxaloacetate	Flow of electrons down ETC
D	Formation of ATP via substrate level phosphorylation	Oxidation of succinate to fumarate	Oxidative phosphorylation

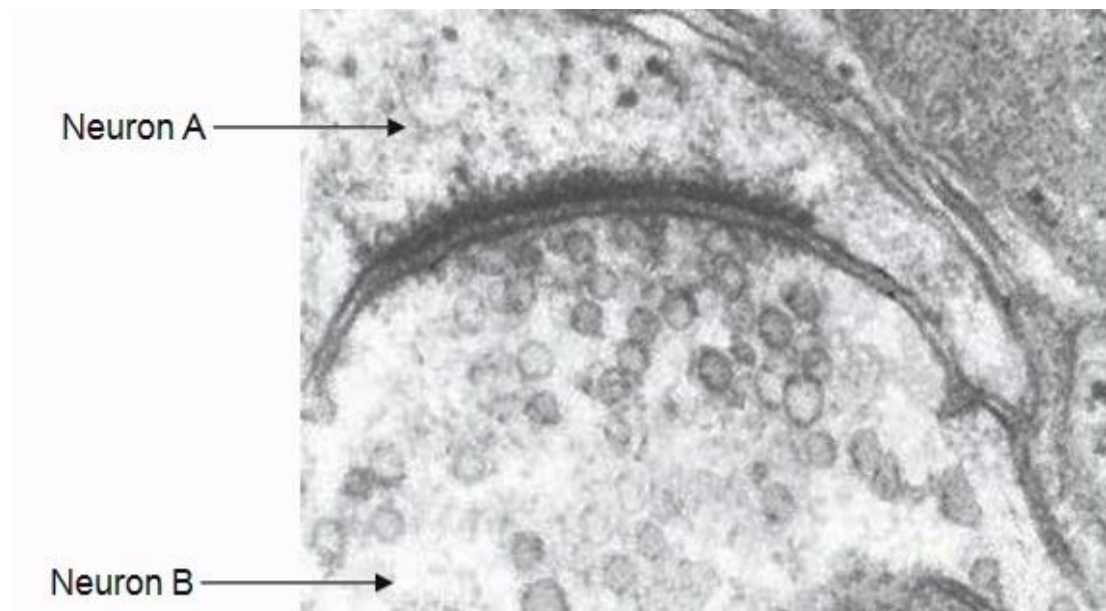
27. In alcoholic fermentation, the final acceptor of electrons stripped from glucose is

- A oxygen
- B ethanol
- C pyruvate
- D acetaldehyde

28. What is necessary for an organism to respond to changes in its internal environment?

- A a communication system between receptors and effectors
- B a communication system involving nerve cells
- C negative feedback from receptor to effector
- D positive feedback from effector to receptor

29. The micrograph below shows a synapse between two neurons.



Which statement about synaptic transmission is correct?

- A Neuron B has received the synaptic vesicle from Neuron A.
- B Calcium ion influx is triggered by depolarisation that is produced by the arrival of an action potential in Neuron A.
- C Neurotransmitters bind to voltage-gated sodium ion channels found in Neuron B, resulting in the generation of a post-synaptic potential.
- D Membrane of Neuron A has many neurotransmitter channels.

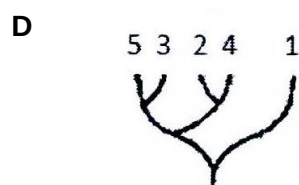
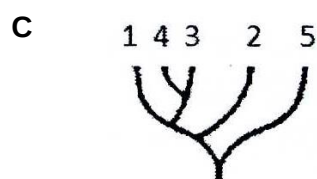
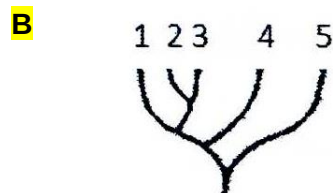
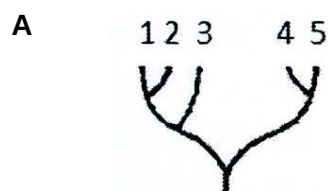
30. Which of the following is a similarity between G protein coupled receptors and receptors tyrosine kinase?

- A Conformational change that results from binding of ligand molecule.
- B Formation of a dimer following binding of a signal molecule.
- C Ligand-binding sites are specific for steroid hormones.
- D Channel proteins that span across the plasma membrane.

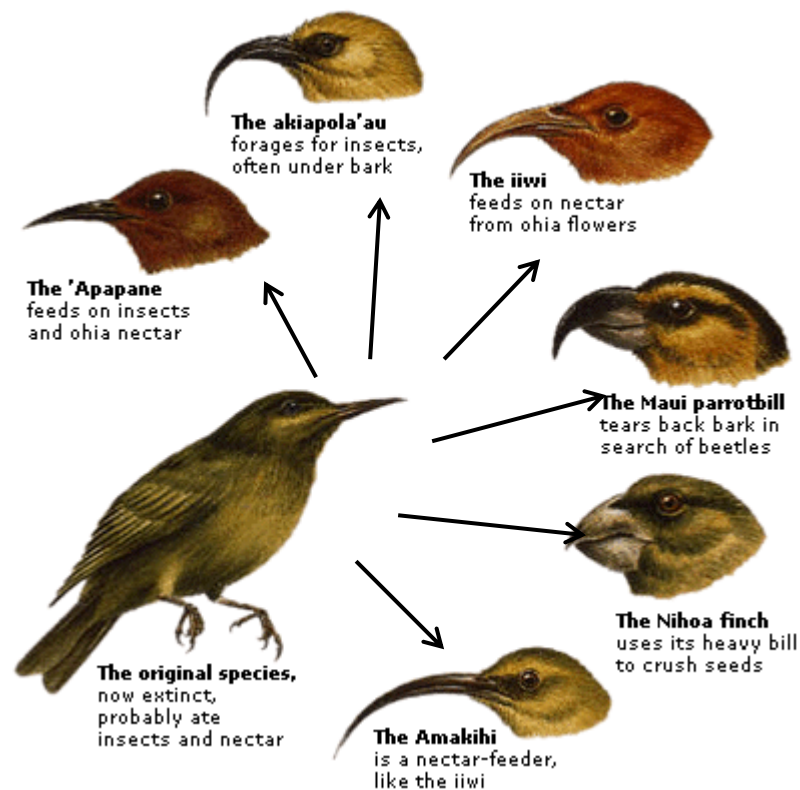
31. Five species 1, 2, 3, 4 and 5 possess the enzyme cytochrome c oxidase. The primary structure of the enzyme was determined for each species and the number of amino acid differences between species is given below.

	1	2	3	4	5
1	0	8	9	21	24
2	8	0	4	20	23
3	9	4	0	19	22
4	21	20	19	0	21
5	24	23	22	21	0

From the above evidence, which one of the following represents the probable evolutionary relationships between the species?



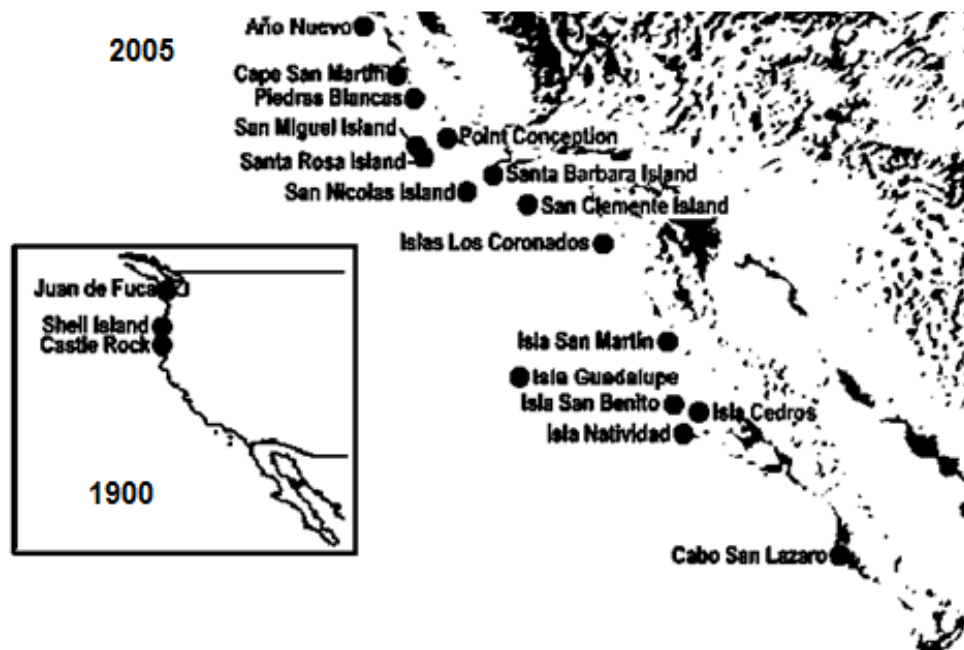
32. The diagram illustrates variation in Honeycreepers of the Family *Drepanididae* from the Hawaiian Islands for a variety of beak shape, size and in eating habits.



How do these examples support Darwin's theory of natural selection?

- A They illustrate how biotic factors operating on a variable genotype can bring about evolutionary change.
- B They illustrate how analogous structures have become adapted to perform different functions.
- C They illustrate how groups with separate evolutionary origins differ over time as a result of divergent evolution.
- D They illustrate the principle of adaptive radiation suggesting the divergent evolution of homologous structures.

33. The northern elephant seal almost became extinct in the late 1800's following harvesting by whalers and sealers for their blubber which contains oil. A small colony of between 20 and 100 individuals survived on Guadalupe Island off Baja California. From that small colony have come the approximately 160,000 elephant seals found on the Pacific coast today as seen in diagram below.



Distribution of northern elephant seal breeding colonies then and now, at 2005 and 1900s.

Why is the degree of homozygosity higher in their descendents than in the original population of northern elephant seal in 1800s?

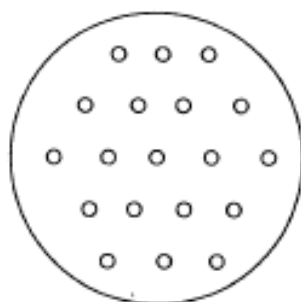
- A Directional selection
- B Genetic bottleneck**
- C Founder effect
- D Genetic drift

34. Molecular homology of haemoglobin is used to determine evolutionary relatedness in a range of mammals. Either nucleic acids or proteins can be sequenced. The sequence of nucleic acids and proteins are compared.

Which sequence will vary the least in related mammals?

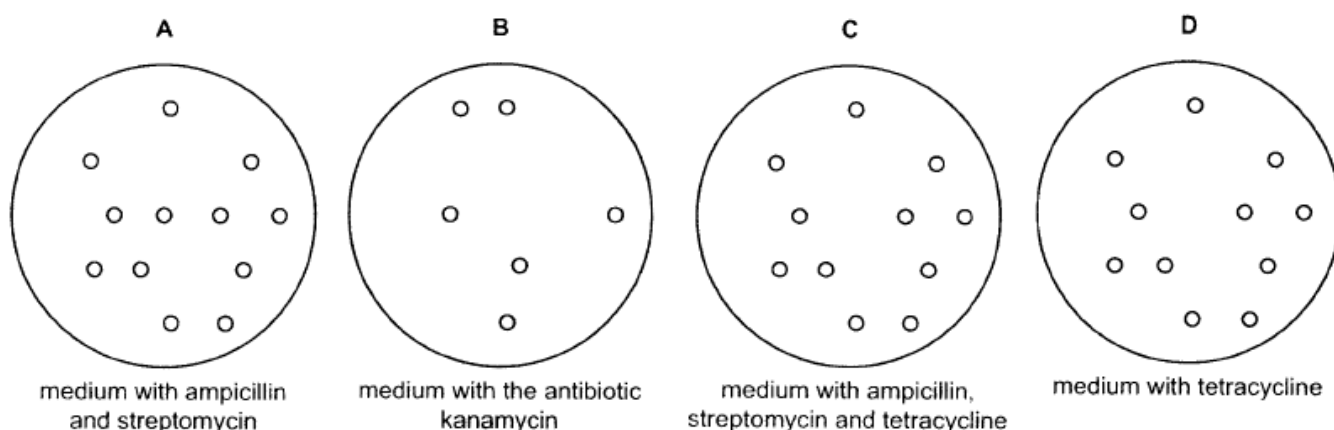
- A Amino acid sequence**
- B DNA
- C Edited mRNA transcript
- D Primary mRNA transcript

35. A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coli* and the bacteria grown on a nutrient medium. The resulting master plate is shown in the diagram.



Transformed cells were selected by replica plating the bacteria colonies onto media containing various antibiotics.

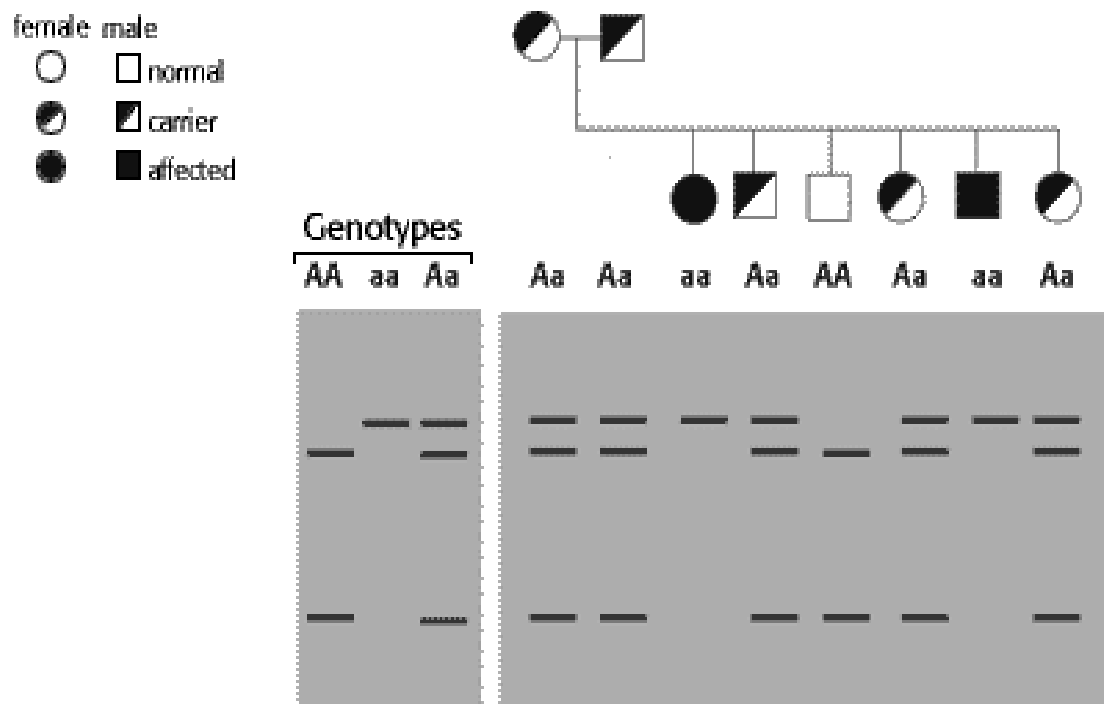
Which plate contains the colony of bacteria into which the human gene has been successfully inserted? **A**



36. The construction of a cDNA library involves a number of stages. What is the correct sequence of stages?

	mRNA is isolated from cells	inserted into a vector and cloned	mRNA is converted into single-stranded cDNA	single-stranded cDNA is hybridized with a primer
A	1	2	4	3
B	2	1	3	4
C	3	4	2	1
D	1	4	2	3

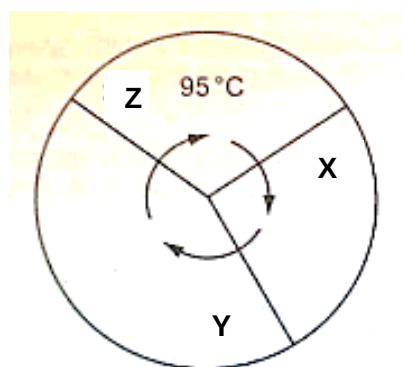
37. Sickle-cell anemia is an autosomal, recessive human disease. A hypothetical pedigree for parents, each heterozygous for the sickle-cell allele, is shown in the figure below. The inheritance of the normal and sickle-cell allele in this family can be followed by RFLP analysis. Each of the different genotypes, homozygous normal (**AA**), heterozygous carrier (**Aa**), and homozygous recessive with sickle-cell anemia (**aa**) can be identified by RFLP analysis.



Based on the above analysis, which of the following statements is true?

- A The sickle-cell anaemia is caused by a mutation resulting in a gain of restriction site.
- B The sick-cell anaemia is caused by a mutation resulting in a loss of restriction site.**
- C The normal allele is cut using a different enzyme from that of the sickle-cell allele, hence resulting in different number of fragments produced.
- D The normal allele is a result of cutting at more than three restriction sites.

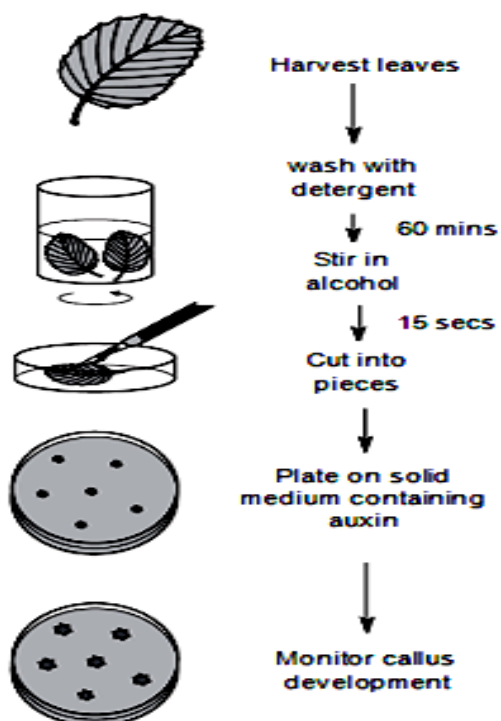
38. The diagram represents the stages of a polymerase chain reaction.



Which row shows the events happening at stages 1, 2 and 3?

	X	Y	Z
A	Binding of primers at 55°C	Polymerisation at 72°C	Denaturing of DNA
B	Denaturing of DNA at 72°C	Binding of primers at 55°C	Polymerisation
C	Polymerisation at 55°C	Binding of primers at 72°C	Denaturing of DNA
D	Polymerisation at 55°C	Denaturing of DNA at 72°C	Binding of primers

39. Callus tissue is produced using leaves by the technique shown in the diagram below. Using this, a student investigated the effect of two different concentrations of auxin on the volume of callus material produced. The table shows the results obtained over five days.



Time (day)	volume of callus material (cm ³)	
	low concentration of auxin added	high concentration of auxin added
0	1.0	1.0
1	1.3	1.7
2	1.9	4.1
3	3.0	8.3
4	3.9	10.6
5	5.4	12.9

Which of the following statements about the experiment is false?

- I. The leaves are disinfected and sterilized by washing with detergent and alcohol.
- II. The sterilization process can alternatively be done using an autoclave machine.
- III. The only nutrient required to stimulate callus formation is auxin.
- IV. Continued subculturing at three- to four-week intervals of small cell clusters taken from calli can maintain the callus cultures for long periods of time.

- A II only
- B I and IV only
- C III only
- D II and III only

40. Some of the limitations, ethical and social concern/s of gene therapy are listed below.

- 1. The definition of normal and disability lies in the decision of a group of scientists.
- 2. Multiple rounds of gene therapy may be required as vectors do not integrate normal allele into host genome.
- 3. Viral vectors have known to produce harmful side effects.
- 4. The presence of a cure might demean the lives of individuals presently affected by disabilities.
- 5. The formation of class distinctions when gene therapy is only available to individuals with financial means.
- 6. Modification of germlines affects subsequent generations.

Which row identified the limitations and the types of concerns?

	Limitations	Ethical concern(s)	Social concern(s)
A	1 and 3	2, 5 and 6	4
B	2 and 3	1, 4 and 6	5
C	2 and 4	1 and 6	3 and 5
D	1, 2 and 3	4 only	5 and 6

PJC Prelims Paper 1 Answers

1	B	11	D	21	C	31	B
2	A	12	C	22	C	32	D
3	C	13	B	23	B	33	B
4	B	14	B	24	A	34	A
5	A	15	B	25	D	35	A
6	B	16	B	26	D	36	D
7	A	17	A	27	D	37	B
8	D	18	D	28	A	38	A
9	A	19	B	29	D	39	D
10	A	20	B	30	A	40	B